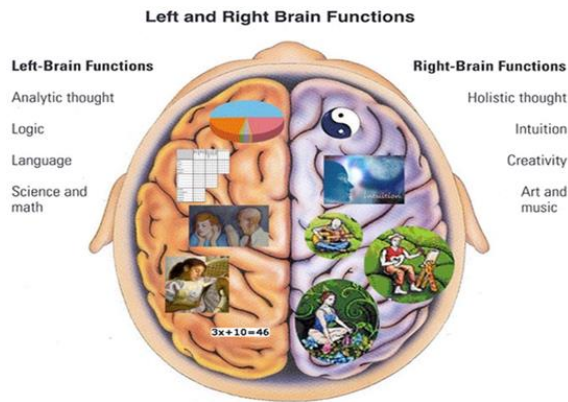




NCM 3116 (Care of Clients with Problems in Nutrition and Gastro-intestinal metabolism and Endocrine Function, Perception and Coordination-Acute and Chronic – LECTURE)

NURSING CARE OF CLIENT WITH NEUROLOGIC PROBLEMS

I. THE HUMAN BRAIN



II. MAJOR DIVISION OF THE NERVOUS SYSTEM

- A. **CENTRAL NERVOUS SYSTEM (CNS)**, consisting of the brain and spinal cord,
- B. **PERIPHERAL NERVOUS SYSTEM (PNS)**, the vast network of spinal and cranial nerves linking the body to the brain and spinal cord.
 - The PNS is subdivided into:
 - AUTONOMIC NERVOUS SYSTEM

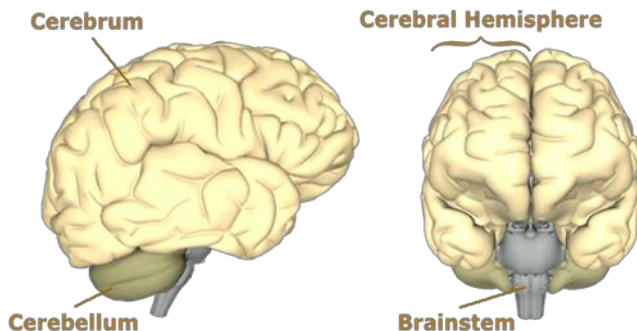
Involuntary control of:

- Internal organs
- Blood vessels
- Smooth and cardiac muscles
- a. SYMPATHETIC NERVOUS SYSTEM
- b. PARASYMPATHETIC NERVOUS SYSTEM
 - SOMATIC NERVOUS SYSTEM

Voluntary control of:

- Skin
- Bones
- Joints
- Skeletal muscles

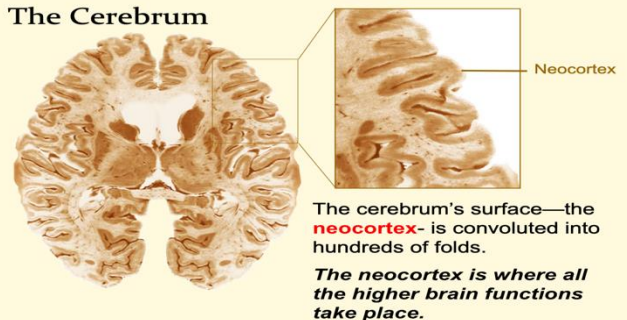
III. EXTERNAL BRAIN STRUCTURES



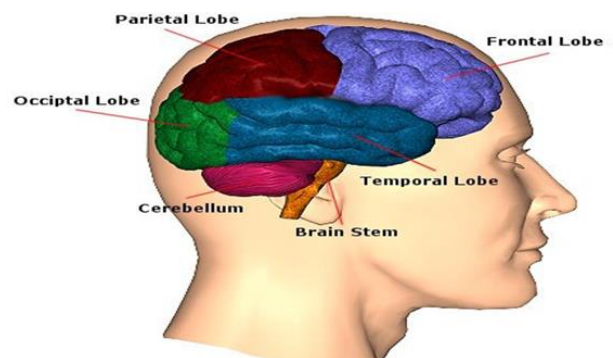
CEREBRUM

- The largest portion of the brain is the cerebrum. It consists of two hemispheres that are connected together at the *corpus callosum*.
- The cerebrum is often divided into **five lobes** that are responsible for different brain functions.

The Cerebrum



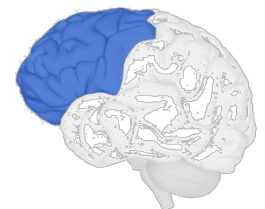
IV. LOBES OF THE CEREBRUM



FRONTAL LOBE

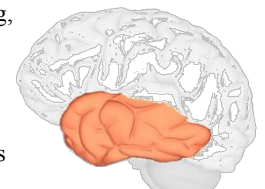
The frontal lobe is the area of the brain responsible for higher cognitive functions. These includes:

- Problem solving
- Spontaneity
- Memory
- Language
- Motivation
- Judgment
- Impulse control
- Social and sexual behavior.



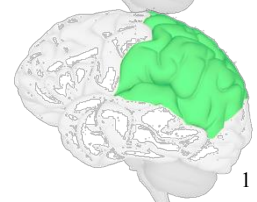
TEMPORAL LOBE

- The temporal lobe plays a role in emotions, and is also responsible for smelling, tasting, perception, memory, understanding music, aggressiveness, and sexual behavior.
- The temporal lobe also contains the language area of the brain.



PARIETAL LOBE

- The parietal lobe plays a role in our sensations of touch, smell,

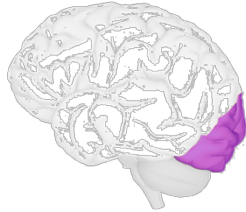


and taste. It also processes sensory and spatial awareness, and is a key component in eye-hand coordination and arm movement.

- The parietal lobe also contains a specialized area called Wernicke's area that is responsible for matching written words with the sound of spoken speech.

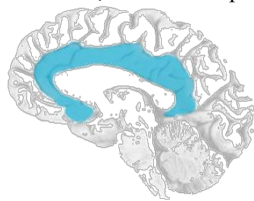
OCCIPITAL LOBE

- The occipital lobe is at the rear of the brain and controls vision and recognition.



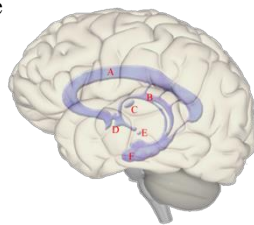
LIMBIC LOBE

- The limbic lobe is located deep in the brain, and makes up the limbic system.



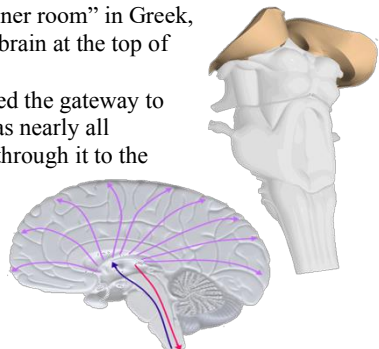
LIMBIC SYSTEM

- The limbic system is the area of the brain that regulates emotion and memory.
- It directly connects the lower and higher brain functions.



THALAMUS

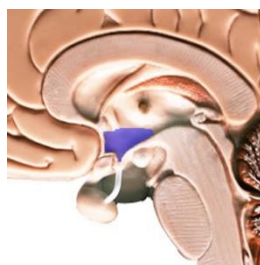
- Thalamus means "inner room" in Greek, as it sits deep in the brain at the top of the brainstem.
- The thalamus is called the gateway to the cerebral cortex, as nearly all sensory inputs pass through it to the higher levels of the brain.



HYPOTHALAMUS

The hypothalamus sits under the thalamus at the top of the brainstem. It controls many critical bodily functions:

- Controls autonomic nervous system
- Center for emotional response and behavior
- Regulates body temperature
- Regulates food intake
- Regulates water balance and thirst
- Controls sleep-wake cycles
- Controls endocrine system



The hypothalamus is shaded blue. The pituitary gland extends from the hypothalamus.

V. BRAIN STEM (EXTENSION OF THE SPINAL CORD)

A. Medulla Oblongata

- Contains Cardiac, Respiratory, Vomiting, and Vasomotor centers (heart rate, respiration, blood vessel diameter, sneezing, vomiting, swallowing, coughing)

B. Pons

- Respiratory center

C. Midbrain

- Responsible for motor coordination
- Contains the visual reflex and auditory relay centers

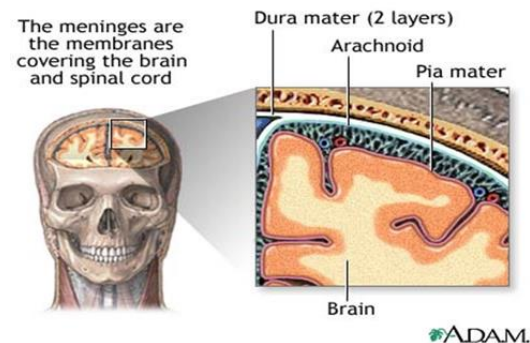
VI. STRUCTURES PROTECTING THE BRAIN

MENINGES

- Covers the brain
- Provides protection, support, and nourishment to the brain and spinal cord.

LAYERS:

1. Dura Mater
2. Arachnoid – Contains the choroid plexus (CSF Production)
3. Pia Mater



CEREBRO SPINAL FLUID

- A clear and colorless fluid produced in the choroid plexus at the lateral 3rd and 4th ventricles.
- Contains minimal WBC and no RBC

CSF CIRCULATION / PATHWAY

- Is produced in the **choroid plexus** (approx. 50 – 70%) and the remainder is formed around blood vessels and along ventricular walls.
- It circulates from the lateral ventricles to the foramen of Monro.
- Third Ventricle, aqueduct of Sylvius (Cerebral aqueduct), Fourth ventricle, foramen of magendie (Median aperture) and foramina of Luschka (lateral apertures)

- Subarachnoid Space over the brain and spinal cord. CSF is reabsorbed into venous sinus blood via arachnoid granulations.

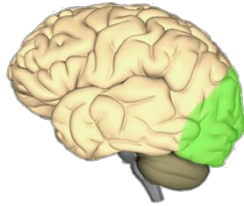
BLOOD – BRAIN BARRIER

- Formed by the endothelial cells of the brain capillaries.
- All substances entering the CSF must filter through these barriers.
- Dyes, medications and antibiotics are examples of substances that cannot reach the CNS.

VII. BRAIN FUNCTIONS

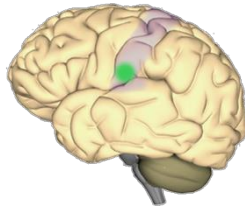
VISION

- The visual cortex resides in the occipital lobe of the brain.
- Sensory impulses travel from the eyes via the optic nerve to the visual cortex.
- Damage to the visual cortex can result in blindness.



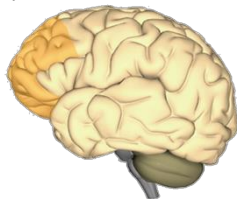
TASTE

- The gustatory complex (green circle) is the part of the sensory cortex (purple area) that is responsible for taste.



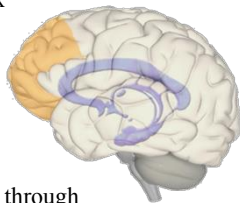
COGNITION

- The prefrontal cortex is involved with intellect, complex learning, and personality.
- Injuries to the front lobe can cause mental and personality changes.



EMOTION

- Emotions are an extremely complex brain function. The emotional core of the brain is the limbic system.
- This is where senses and awareness are first processed in the brain.
- Mood and personality are mediated through the prefrontal cortex. This part of the brain is the center of higher cognitive and emotional functions.
- **YELLOW:** PREFRONTAL CORTEX
- **PURPLE:** LIMBIC SYSTEM



SPEECH

- Broca's area is where we formulate speech and the area of the brain that

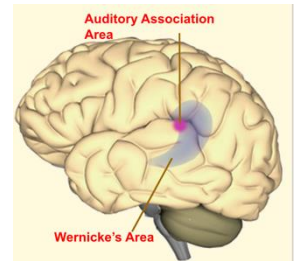


sends motor instructions to the motor cortex

- Injury to Broca's area can cause difficulty in speaking. The individual may know what words he or she wishes to speak, but will be unable to do so.
- **PINK:** BROCA'S AREA

LANGUAGE

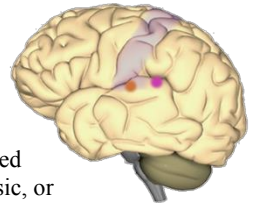
- Wernicke's area is a specialized portion of the parietal lobe that recognizes and understands written and spoken language.
- Wernicke's area surrounds the auditory association area.
- Damage to this part of the brain can result in someone hearing speech, but not understanding it.



HEARING

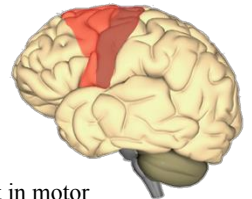
There are two auditory areas of the brain:

- The primary auditory area (brown circle) is what detects sounds that are transmitted from the ear. It is located in the sensory cortex.
- The auditory association area (purple circle) is the part of the brain that is used to recognize the sounds as speech, music, or noise.



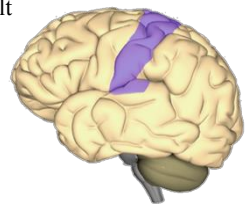
MOTOR CORTEX

- The light red area is the pre-motor cortex, which is responsible for repetitive motions of learned motor skills. The dark red area is the primary motor area, and is responsible for control of skeletal muscles.
- Injury to the motor cortex can result in motor disturbance in the associated body part.

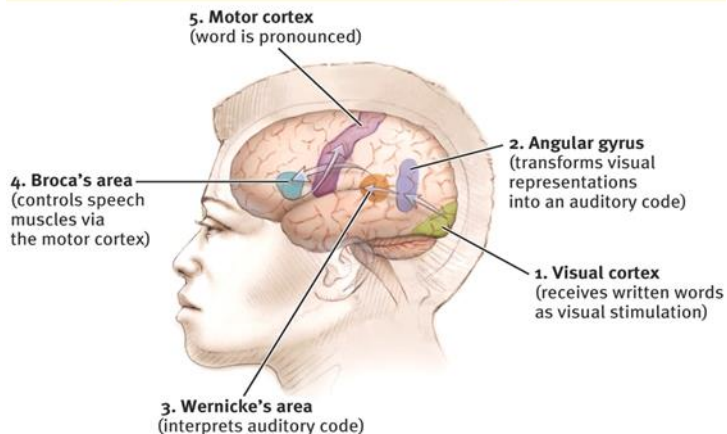


SENSORY CORTEX

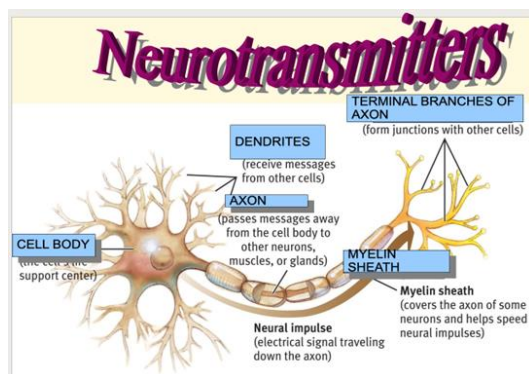
- Injury to the sensory cortex can result in sensory disturbance in the associated body part.



Specialization and Integration



VIII. NEUROTRANSMITTERS



NEURAL COMMUNICATION

SYNAPSE (SIN – aps)

- Junction between the axon tip of the sending neuron and the dendrite or cell body of the receiving neuron
- Tiny gap at this junction is called the synaptic gap or cleft

NEUROTRANSMITTERS

- Chemical messengers that traverse the synaptic gaps between neurons
- When released by the sending neuron, neuro-transmitters travel across the synapse and bind to receptor sites on the receiving neuron, thereby influencing whether it will generate a neural impulse

NEUROTRANSMITTERS

- There are approximately 50 neurotransmitters identified
- Are chemicals located and released in the brain to allow an impulse from one nerve cell to pass to another nerve cell.
- Nerve cells communicate messages by secreting neurotransmitters.
- Neurotransmitters can excite or inhibit neurons

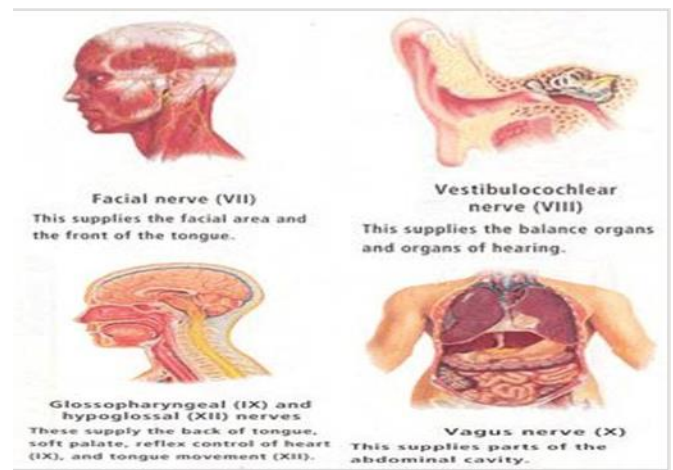
IX. CRANIAL NERVES

CRANIAL NERVES

Are composed of twelve pairs of nerves that emanate from the nervous tissue of the brain.

NERVES in ORDER	MODALITY	FUNCTION
I. Olfactory	Special Sensory	Smell
II. Optic	Special Sensory	Vision
III. Oculomotor	Somatic Motor	Levator palpebrae, superioris, superior, medial & inferior recti muscles
	Visceral Motor	Parasympathetic to ciliary & pupillary constrictor muscles
IV. Trochlear	Somatic	Superior oblique muscle
V. Trigeminal	Branchial Motor	Muscles of mastication
	General Sensory	Sensory for head/neck, sinuses, meninges, & external surface of tympanic membrane
VI. Abducens	Somatic Motor	Lateral rectus muscles
VII. Facial	Branchial Motor	Muscles of facial Expression
	Visceral Motor	Parasympathetic to all glands of head except the parotid
	General Sensory	Sensory for ear and tympanic membrane
VIII. Vestibocochlear / Auditory Nerve	Special Sensory	Taste anterior two – thirds of tongue
	Special sensory	Hearing and balance
IX. Glossopharyngeal	Brachial motor	Stylopharyngeus muscles
	Visceral motor	Parotid gland
	Visceral sensory	Carotid body
	General sensory	Sensation posterior one third tongue & internal surface of tympanic membrane
	Special	Taste posterior one

		sensory	– third tongue
X. Vagal	Brachial Motor		Muscles pharynx & larynx
	Visceral Motor		Parasympathetic to neck, thorax, & abdomen
	Visceral Sensory		Sensory from pharynx, larynx & viscera
	Special Sensory		Sensory from external ear
XI. Accessory nerve / Spinal Accessory	Branchial Motor		Trapezius & Sternocleidomastoid muscles
XII. Hypoglossal	Somatic Motor		Tongue muscles except palatoglossal

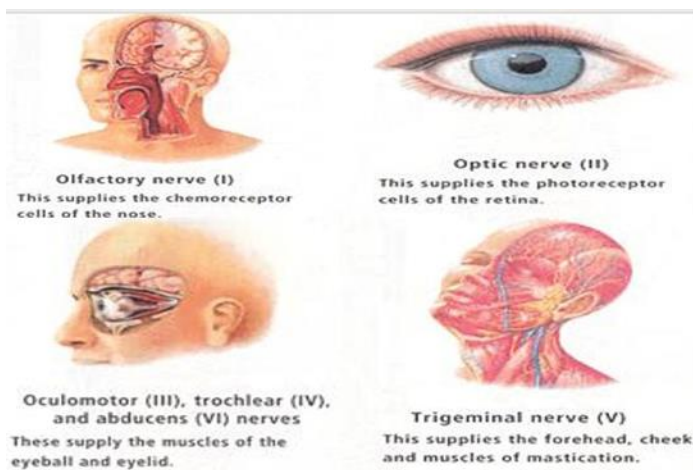
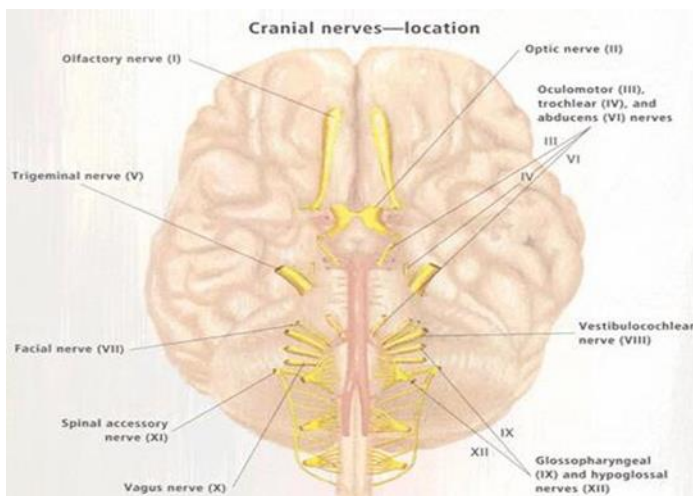


X. NEURODIAGNOSTIC STUDIES

LUMBAR PUNCTURE/SPINAL TAP

- for CSF analysis.
- CSF should be clear and colorless
- Obtained for cell count, culture, glucose and protein testing
- Also used to measure CSF fluid or pressure, or instill air, dye or meds
- A spinal needle is inserted into the subarachnoid space between 3rd and 4th or 4th and 5th lumbar vertebrae

CRANIAL NERVES LOCATION



NURSING MANAGEMENT IN ASSISTING WITH SPINAL TAP / LUMBAR PROCEDURE

Pre Procedure

- Signed consent
- Explain procedure and clarify misconceptions

During the Procedure

- Position: Back of patient towards MD, lateral recumbent has client draw knees to abdomen and chin to chest.
- Pillows on patient's head and between legs.
- Maintain position to prevent trauma resulting to a bloody tap.
- Instruct patient to breathe normally and relax; hyperventilation may lower an elevated pressure altering accurate pressure reading.
- Assist in cleansing lumbar area with betadine solution and draping (observe strict aseptic technique).
- MD will inject a local anesthetic to numb puncture site.
- A specimen is collected usually in 3 tubes; MUST be labeled in order of collection. Pressure reading may also be obtained.

Post Procedure

- Patient must lie PRONE for 2-3 hours to reduce leakage of CSF.
- Monitor for complications of Lumbar tap.
- Force fluid intake to prevent post-procedure headaches.
- Monitor intake and output.

POST LUMBAR TAP COMPLICATIONS

- Post Lumbar Puncture Headache
- Temporary voiding problems
- Slight elevation of temperature
- Backache or spasms
- Stiff neck

- Rare but serious: herniation of spinal intracranial contents, spinal epidural abscess, meningitis

RADIOGRAPHIC

1. Skull and Spinal Radiography

- X-ray of the skull reveal the size and shape of the skull bones, suture separation in infants, fracture or bony defects, erosion or calcification.

2. CT Scan

- Images provide cross-sectional view of the brain, distinguish differences in tissue densities of the skull, cortex, sub cortical structures and ventricles.
- Image is displayed on an oscilloscope or TV monitor and is photographed digitally.
- Performed FIRST without contrast before imaging with contrast.

CT SCAN NURSING INTERVENTIONS

- Patient needs to lie still during the entire procedure; essential to be instructed to patient.
- Teach relaxation techniques.
- Sedation maybe necessary if patient is confused, agitated or restless.
- Will require monitoring all through out.

If a contrast agent/dye will be used:

- Assess allergy to iodine/shellfish
- 4hour fasting is required prior to test
- Monitor post procedure dye reaction such as vomiting, nausea, flushing
- Assess dye injection site for bleeding or hematoma, and monitor presence of distal pulses
- Provide fluid replacement because diuresis from dye is expected

3. Magnetic Resonance Imaging

- Uses magnetic field to create an image
- Provides info about chemical changes w/in the cell to monitor an organ's response to treatment
- Maybe used with or without dye

MRI NURSING INTERVENTIONS

- Assess patient for claustrophobia.
- Patient instructions: relaxation technique; microphone is provided inside the scanner so patient may speak to staff.
- All metal objects, credit cards, med patches with metal/foil backing (causes burns) must be removed.
- No metal objects, such as o2 tanks, maybe brought inside.

XI. CEREBRAL ANGIO GRAM

- Injection of contrast through the femoral artery into carotid arteries to visualize arteries and assess for lesions.
- Used to investigate vascular disease, aneurysms.
- Frequently performed before craniotomy to assess patency and adequacy of cerebral circulation.
- May use brachial artery to inject dye or a direct puncture of carotid or vertebral artery alternatively.

ANGIOGRAPHY NURSING INTERVENTION

Pre procedure

- Signed consent
- 4-6hours NPO
- Obtain baseline neuro assessment
- Patient must be well hydrated 2days prior to the exam. Patient will be asked to void before going to the x-ray department
- Explain procedure to patient.
- Patient has to be immobile during the entire procedure.
- A brief feeling of warmth in the face, behind the eyes, or the jaw, teeth, tongue, lips will be felt.
- A metallic taste after the injection of the contrast agent will be experienced.
- Mark peripheral pulses

During Procedure

- The groin will be shaved and prepared, injected with a local anesthetic to numb as catheter is inserted
- Catheter will be flushed with heparinized saline

Post Procedure

- Observe for signs of altered cerebral blood flow due to minor or major arterial blockage caused by thrombosis, embolism or hemorrhage producing neurologic deficits such as:
 - a) Altered level of responsiveness and consciousness
 - b) Hemiparesis
 - c) Speech disturbances
- Monitor Neuro status and V/S until stable.
- Monitor for swelling in the neck and for dysphagia and NOTIFY MD if these occur.
- Maintain bed rest for 12hrs as prescribed.
- Elevate head of bed 15-30 degrees only if prescribed.
- Keep bed flat if femoral artery is used, as prescribed.
- Assess peripheral pulses.
- Apply sandbag and pressure dressing to the injection site as prescribed.
- Place ice on the puncture site as prescribed.

XII. SURGICAL INTERVENTION

CRANIOTOMY

- A surgical procedure that involves an incision through the cranium to remove accumulated blood or a tumor.

COMPLICATIONS

- Increased ICP from cerebral edema
- Hemorrhage
- obstruction of the normal flow of CSF

Preoperative Intervention

- Explain procedure to client and family
- Ensure informed consent has been obtained

- Make sure Pre-Operative Clearance is OK as per doctor's order
- Prepare to shave clients head as prescribed

Postoperative Intervention

- Monitor v/s and neuro status q 30mins- 1 hr.
- Monitor increased ICP.
- Monitor for decreased LOC, motor weakness or paralysis, aphasia, visual changes and personality changes.
- Maintain mech vent and slight hyperventilation for the first 48hrs to prevent increase in ICP as per doctor's order.
- Stabilize client before surgery.
- Avoid neck or hip flexion and maintain head in midline neutral position.
- Provide quiet environment.
- Monitor head dressing frequently for signs of drainage
- Mark areas of drainage at least once/shift.
- If on ventriculostomy maintain suction of drain, record amount and color; notify MD if drainage is greater than 30-50ml.
- Record strict measurement of I&O hourly.
- Monitor electrolyte values.
- Apply ice pack or cool compresses as prescribed for periorbital edema and ecchymosis.

Positioning of Patient after Craniotomy

- Varies
- Incorrect positioning may cause serious and possibly fatal complications.

INFRATENTORIAL SURGERY (CEREBELLUM)

- Flat position without head elevation or may order head of bed at 40-35 degrees.
- Do not elevate the head in the acute phase of care after surgery w/o MD order.

SUPRATENTORIAL SURGERY (CEREBRUM)

- Head of bed elevation 30 degrees to promote venous outflow through the jugular vein.
- Do not lower head of bed w/o MDs order.

XIII. COMMON NEUROLOGIC MANIFESTATIONS

HEADACHE

- (Cephalgia)
- Is one of the most common of all human physical complaints.
- Symptom rather than a disease entity.
- It may indicate organic disease (neurologic or other disease), a stress response, vasodilation (migraine), skeletal muscle tension (tension headache), or a combination of factors.

PRIMARY HEADACHE

- Include migraine, tension-type, and cluster headaches
- **Migraine**
 - Is a symptom complex characterized by periodic and recurrent attacks of severe headache.

- Caused by primarily a vascular disturbance that occurs more commonly in women and has a strong familial tendency.

XIV. CLINICAL MANIFESTATION

MIGRAINE

- The migraine with aura can be divided into four phases: prodrome, aura, the headache, and recovery (headache termination and postdrome).

PRODROME

- Symptoms include depression, irritability, feeling cold, food cravings, anorexia, change in activity level, increased urination, diarrhea, or constipation.
- Occur hours to days before a migraine headache.

AURA PHASE

- Characterized by focal neurologic symptoms, visual disturbances (i.e., light flashes and bright spots) are common and may be hemianopic (affecting only half of the visual field).
- Other symptoms that may follow include numbness and tingling of the lips, face, or hands; mild confusion; slight weakness of an extremity; drowsiness; and dizziness.
- Usually lasts less than an hour and may provide enough time for the patient to take the prescribed medication to avert a full-blown attack.

HEADACHE PHASE

- As vasodilation and a decline in serotonin levels occur, a throbbing headache intensifies over several hours.
- This headache is severe and incapacitating and is often associated with photophobia, nausea, and vomiting.
- Its duration varies, ranging from 4 to 72 hours.

RECOVERY PHASE

- In the recovery phase (termination and postdrome), the pain gradually subsides.
- Muscle contraction in the neck and scalp is common, with associated muscle ache and localized tenderness, exhaustion, and mood changes.
- Any physical exertion exacerbates the headache pain.
- During this post headache phase, patients may sleep for extended periods.

TENSION HEADACHE

- Is characterized by a steady, constant feeling of pressure that usually begins in the forehead, temple, or back of the neck.
- It is often bandlike or may be described as "a weight on top of my head."
- Tend to be more chronic than severe and are probably the most common type of headache

CLUSTER HEADACHE

- Severe form of vascular headache.
- Are unilateral and come in clusters of one to eight daily, with excruciating pain localized to the eye and orbit and radiating to the facial and temporal regions.
- The pain is accompanied by watering of the eye and nasal congestion.

- Each attack lasts 30 to 90 minutes and may have a crescendo–decrescendo pattern.
- Often described as penetrating and steady
- They are seen five times more frequently in men than women.

SECONDARY HEADACHE

- Associated with an organic cause, such as a brain tumor or an aneurysm.
- Serious disorders related to headache include brain tumors, subarachnoid hemorrhage, stroke, severe hypertension, meningitis, and head injuries.

Medical Management	Nursing Management
Triptans • Serotonin receptor agonists, antimigraine agents • Cause vasoconstriction, reduce inflammation, and may reduce pain transmission. <u>The five triptans in routine clinical use include:</u> • Sumatriptan (Imitrex) • Naratriptan (Amerge) • Rizatriptan (Maxalt) • Zolmitriptan (Zomig) • Almotriptan	Relieving pain • Comfort measures • Application of local heat or massage. • Additional strategies may include the use of analgesic agents, antidepressant medications, and muscle relaxants as prescribed. • Encourage relaxation techniques such as dbc, guided imagery and etc.

XV. INCREASED INTRACANIAL PRESSURE

- The rigid cranial vault contains brain tissue (1,400 g), blood (75 mL), and CSF (75 mL).
- The volume and pressure of these three components are usually in a state of equilibrium and produce the ICP.
- ICP is usually measured in the lateral ventricles
- It can result from head injury, brain tumor, hydrocephaly, meningitis, encephalitis, or intracerebral hemorrhage.
- Increased ICP should be treated as a medical emergency.

MANIFESTATIONS OF INCREASED ICP

- Headache
- Change in level of consciousness
- Irritability
- Increased systolic BP
- Decreased HR (early)
- Increased HR (late)
- Decreased RR
- Hemiparesis
- Loss of oculomotor control
- Photophobia (light sensitivity)
- Vomiting (with subsequent decreased headache)
- Diplopia (double vision)
- NOTE: BLAP

MANAGEMENT OF INCREASED ICP

1. Elevate head of bed 15 to 30 degrees. Keep head in neutral alignment. Do not flex or rotate neck.
2. Establish IV access.
3. Insert Foley catheter as per doctor's order.
4. Meds that may be used include osmotic diuretics, sedatives, neuromuscular blocking agents, corticosteroids, and anticonvulsants.
5. Restrict fluids follow the doctor's fluid regimen.
6. Closely monitor vital signs and perform neuro checks.
7. Monitor fluids and electrolytes
8. Schedule all procedures (including bathing and especially suctioning) to coincide with periods of sedation.
9. Discourage patient activities that result in use of Valsalva's maneuver.
10. Keep environment as quiet as possible.
11. Ventilator may be used to maintain PaCO₂ between 35-45.

XVI. NEUROLOGICAL DYSFUNCTION

XVII. NEURAL TUBE DEFECT

- Neural tube defects (NTDs) are one of the most common birth defects, occurring in approximately one in 1,000 live births in the United States.
- An NTD is an opening in the spinal cord or brain that occurs very early in human development

ETIOLOGY

- It is UNKNOWN, only associated with the following
- Inadequate folic acid
- Medications – antimetabolites of folic acid, anticonvulsant
- Pregestational IDDM
- Maternal hyperthermia
- Genetics
- Amniotic band disruption sequence
- Teratogens
- Radiation

RISK FACTORS

- Maternal malnutrition
- Drug exposure
- Low socioeconomic group
- Older maternal age

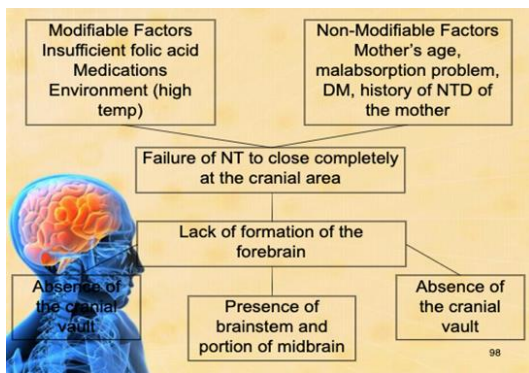
DIAGNOSTICS

- Amniocentesis (to det increased levels of alpha-fetoprotein)
- Alpha-fetoprotein level – increase suggests a NTD)
- Urine estriol level
- Pre-pregnancy serum folic acid
- Fetal ultrasound
- MRI
- CT scan

TYPES

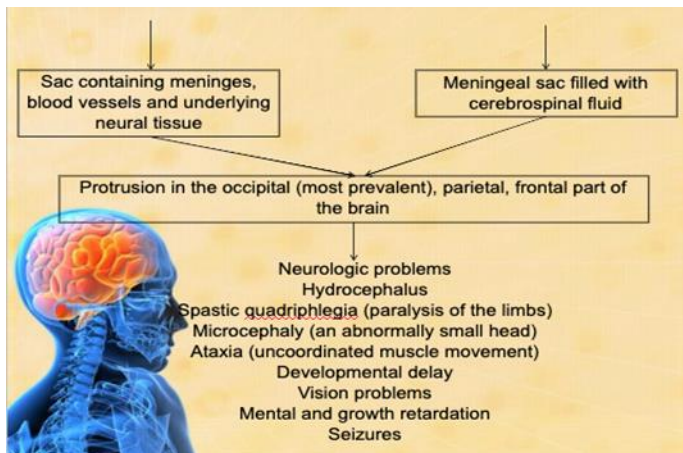
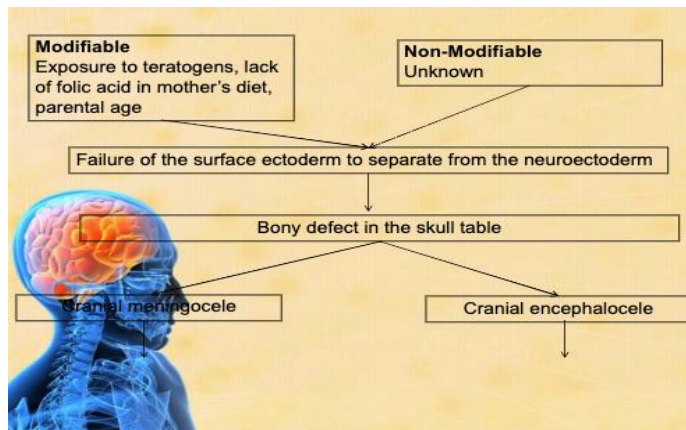
ANENCEPHALY

- Large skull defect with no cortex. many are stillborn and other die within days of birth



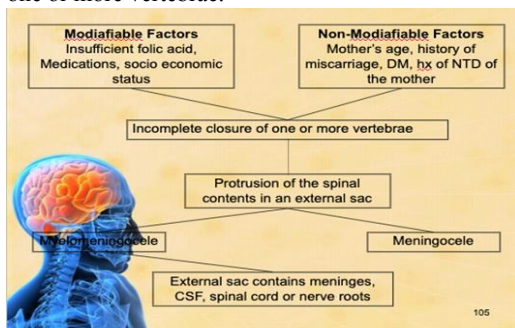
ENCEPHALOCELE

- Projections of cranial content through a bony skull defect, usually occipital region.



SPINA BIFIDA

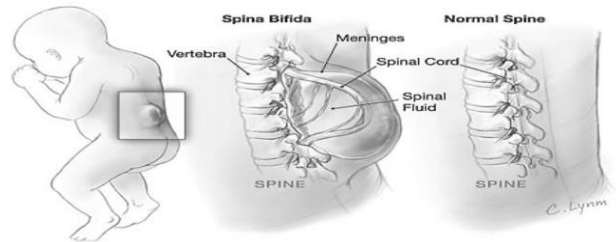
- CNS defect that occurs as a result of incomplete closure of one or more vertebrae.



XVIII. TYPES OF SPINA BIFIDA

SPINA BIFIDA OCCULTA

- Defect in the vertebral arches without herniation of any spinal content.
- Birthmarks, dimples, or hairy tuft at the base of the back.



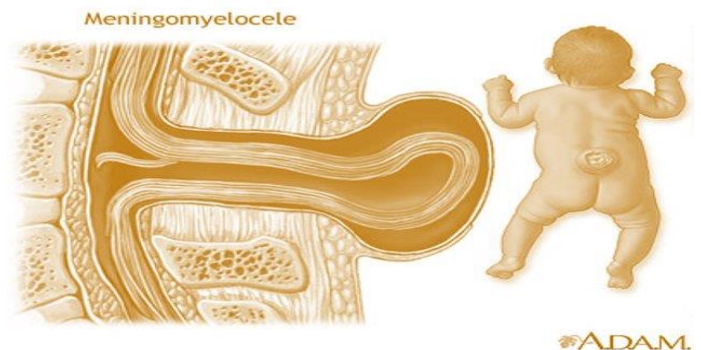
SPINA BIFIDA W/ MENINGOCELE

- Defects in vertebral arches with protrusion of meninges.



SPINA BIFIDA W/ MENINGOMYELOCELE

- Defect of vertebral arches with protrusion of meninges, spinal cord, CSF, nerve roots.



CLINICAL MANIFESTATION

- Depends on the spinal cord involvement
- Visible spinal defect
- Flaccid paralysis of the legs
- Altered bladder and bowel function
- Hip and joint deformities

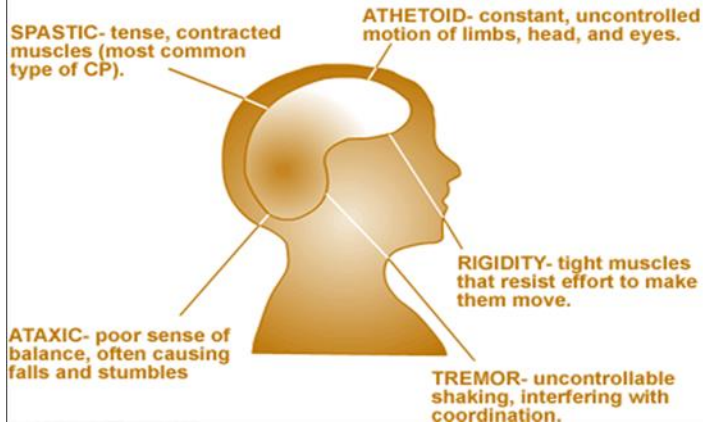
XIX. CEREBRAL PALSY

- Non progressive disorder of movement and posture that results from lesion of the immature brain.

RISK FACTORS

- Prematurity
- Birth asphyxia
- Early infection or trauma

XX. TYPES OF CEREBRAL PALSY



CLINICAL MANIFESTATION

- Tones remains relatively constant regardless of activity and level of arousal
- Significant Hyperreflexia
- Stiff and rigid arms or legs
- Persistence of Primitive and pathologic reflexes
- Extreme irritability and crying
- Feeding difficulties
- Delayed gross development
- Opisthotonos posture

MANAGEMENT

- Multidisciplinary approach
- Therapeutic goal includes minimizing impairments, maximizing function, preserving general health
- Therapies:
 - Speech, physical, occupational
 - Botulinum toxin a: botox
 - Massage
- Prosthetics/Orthotics
- Surgery
- Hyperbaric Oxygen

NOTE:

*** *Injecting botulinum toxin type-A would result in decreased muscle activity by blocking the release of acetylcholine from the neuron*

*** *hyperbaric oxygen therapy (HBOT), is the medical use of oxygen at a level higher than atmospheric pressure*

NURSING MANAGEMENT

- Assess child's developmental level and intelligence
- Encourage early intervention and participation in school programs
- Prepare for using mobilizing devices to prevent deformities
- Communicate with the child on a functional level, not on chronological age level
- Provide safe environment
- Provide safe, appropriate toys for age and developmental level
- Position upright after meals

XXI. REYE'S SYNDROME

- Also known as Reye syndrome, is a rare but serious condition that causes swelling in the liver and brain.
- Reye's syndrome can occur at any age but usually affects children and teenagers after a viral infection, most commonly the flu or chickenpox.

REYE'S SYNDROME CLINICAL STAGING SYSTEM

MILD – MODERATE

STAGE 1	Vomiting, Laboratory evidence of liver dysfunction, lethargic, sleepy.
STAGE 2	Deeply lethargic, restless, confused, delirious, combative, hyperventilation, hyperreflexia
STAGE 3	Obtunded or in a light coma, decorticate rigidity
SEVERE	
STAGE 4	Deepening coma, seizures, decerebrate rigidity, fixed pupils, loss of oculovestibular reflexes.
STAGE 5	Seizures, deep coma, flaccid paralysis, absent deep tendon reflexes, respiratory arrest, and fixed, dilated pupils

- The use of aspirin during a viral illness has most commonly been linked to Reye's syndrome.
- Medium-chain acyl-CoA dehydrogenase (MCAD) deficiency. MCAD deficiency is a fatty acid oxidation disorder. Fatty acid oxidation disorders are a group of inherited metabolic disorders in which the body is unable to break down fatty acids. This happens because an enzyme is missing or not working properly. In people with a fatty acid oxidation disorder, aspirin use during a viral illness is more likely to trigger symptoms of Reye's syndrome. A screening test can determine if your child has a fatty acid oxidation disorder.

CDC DIAGNOSTIC CRITERIA FOR REYE'S SYNDROME

Acute noninflammatory encephalopathy with an altered level of consciousness.
Hepatic dysfunction with a liver biopsy showing fatty metamorphosis or a more than 3 – fold increase in alanine aminotransferase (ALT), aspartate aminotransferase (AST), and/or ammonia levels.
No other explanation for cerebral edema or hepatic abnormality.
Cerebral Spinal Fluid (CSF) with white blood cells (WBC) 8/mm3 or fewer.
Brain biopsy showing cerebral edema without inflammation.

TREATMENT

- No specific treatment
- Supportive care
- 10% dextrose,
- Blood transfusion
- intubation may be necessary to provide adequate oxygenation
- Intracranial pressure must be monitored and controlled in patients with cerebral edema.
- Management of cerebral edema may involve fluid restriction, diuresis, and corticosteroids. Pentobarbital has been given to patients to decrease cerebral metabolic demands.²³ Therapies to reduce hyperammonemia may also be indicated. In patients

who develop seizures, antiepileptic medications may be necessary.

XXII. MULTIPLE SCLEROSIS

- Immune-mediated progressive demyelinating (myelin sheath of neurons is damaged) disease of the CNS
- A chronic degenerative progressive disease of the CNS
- Characterized by small patches of demyelination in the brain & spinal cord.
- Aka: disseminated sclerosis

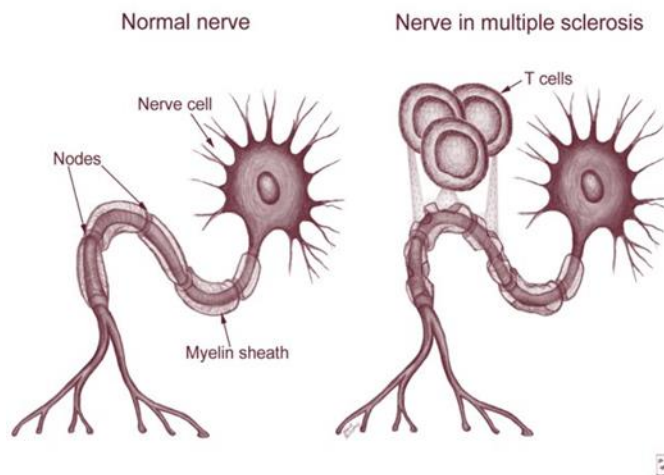
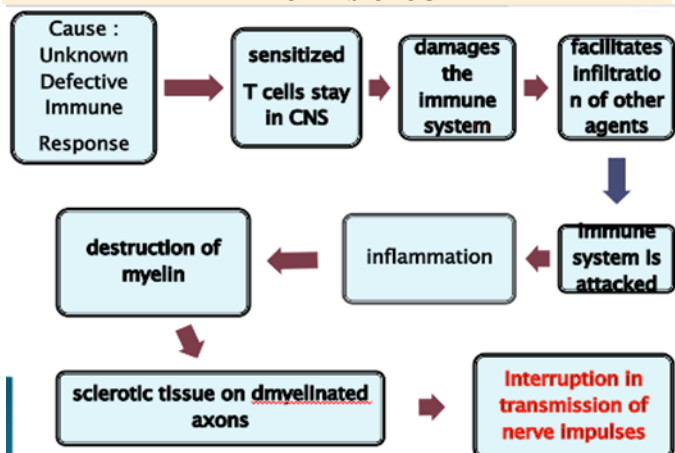
EPIDEMIOLOGY

- Major cause of chronic disability in young adults
- Onset of symptoms usually occurs between 20 and 40.
- Occurs more frequently in Caucasian females than males

ETIOLOGY

- Cause is not known
- Current theories identify autoimmune- mediated inflammatory demyelination and axonal injury
- Multiple factors are involved (research)

PATHOPHYSIOLOGY



CLINICAL FINDINGS

Ent	Musculo
• Nystagmus • Diplopia • Blindness • Blurred vision • Pallor on optic disc evident upon eye exam (ophthalmoscope)	• paresthesia • altered position Sense • ataxia • intention tremors • spastic paralysis & gait • increased deep tendon reflex

Neuro	Other signs
• Inappropriate emotional affect • Slurred or <u>scanning speech</u> • Cognitive loss • Increased gamma globulin levels in the CSF	• dysphagia • Impaired bowel & <u>bladder function</u>

DIAGNOSTIC METHODS

A. MRI

- To visualize small plaques, evaluate cause and effects of treatment.

B. Electrophoresis

- Used to study the CSF, abnormal IgG appears in the CSF
- A technique widely used to separate and identify serum proteins and other substances up to 95% of patient.

C. Fluid – Attenuated – Inversion – Recovery (FLAIR)

- Able to detect two or three times the number of lesions seen in MRI

D. Magnetic Resonance Spectroscopy

- Decreased level of N – Acetyl aspartate (NAA) suggest presence of axonal damage
- N – Acetyspartate – is the second most concentrated molecule in the brain after the amino acid glutamate.

MEDICAL MANAGEMENT

Immunomodulating Drugs – slows disease progression, reduces relapse.

- E.g., Interferon beta – 1a (Avonex)

Immunosuppressant Drugs – same as # 1

- Mitoxantrone (Novantrone)

Corticosteroid Drugs – exacerbations

- Corticotropin (ACTH), Prednisone Deltasone

Muscle Relaxants – spasticity

- Diazepam (Valium), Baclofen (Lioresal)

Antiepileptic Drugs – neuropathic pain

- Carbamazepine (Tegretol)

Anti-depressants – depression

- Amitriptyline (Elavil)

Stimulant Drugs – fatigue

- Amantadine (Symmetrel)

Cholinergic – urinary retention

- Neostigmine (Prostigmin)

Anti cholinergic – urinary frequency

- Probanthin (Pro-banthine)

- Enhancing bladder and bowel control
- Enhancing communication
- Managing swallowing difficulties
- Suction apparatus
- Evaluate sac and measure lesion
- Perform neurological assessment
- Monitor increase in ICP
- Protect sac, using aseptic technique
- Place in prone position, head turned to one side for feeding
- Assess for early signs of infection
- Administer medication as ordered (antibiotics)

XXIII. TRIGEMINAL NEURALGIA

- **TIC DOULOUREUX**
- Is a condition of fifth cranial nerve characterized by paroxysms of pain in the area innervated by any of the three branches.
- Most commonly occurs in the second and third branches of the trigeminal nerve.

CAUSES / RISK FACTORS

- Demyelination of axons in the ganglion, root and nerve.
- Multiple sclerosis
- Swollen blood vessels or tumor
- More common in women between 5th and 6th decade of life

MANIFESTATIONS

- Pain that ends as abruptly as it starts and is described as a unilateral shooting and stabbing sensation.
- Associated involuntary contraction of the facial muscles can cause sudden closing of the eye or a twitch of the mouth.

Factors that trigger:

- Paroxysms can occur with any stimulation of the terminals of the affected nerve branches such as *washing the face, shaving, brushing the teeth, eating, and drinking.*

MEDICAL MANAGEMENT

PHARMACOLOGIC THERAPY

1. **Anticonvulsant agents** – such as carbamazepine (Tegretol)
 - Relieve pain by reducing the transmission of impulses at certain nerve terminals.
 - S/E: nausea, dizziness, drowsiness, and aplastic anemia.
 - Monitor for bone marrow depression during long term therapy.
2. **Gabapentin (Neurontin) and baclofen (Lioresal)** – used for pain control.
3. **Phenytoin (Dilantin)** – may be used as adjunctive therapy.

SURGICAL MANAGEMENT



Administration of Lioresal through a catheter and a pump



Intrathecal (area surrounding the spinal cord) medication delivery

NURSING MANAGEMENT

- Regular exercise
- Warm packs
- Muscle stretching
- Stretch – hold – relax routine, swimming, stationary bicycling, progressive weight bearing
- Activity and rest
- Minimizing effects of immobility
- Safety

SURGICAL MANAGEMENT

Microvascular Decompression of the Trigeminal Nerve

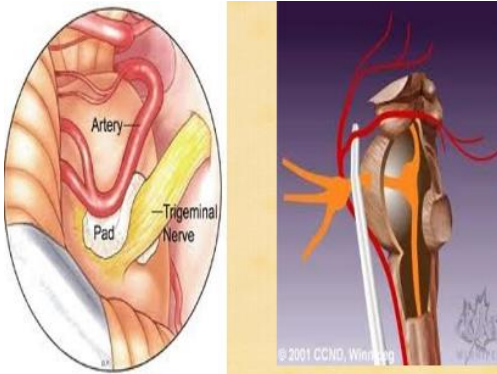
- Intracranial approach used to decompress the trigeminal nerve by relieving the contact between the cerebral vessel and trigeminal nerve root entry.

Percutaneous Radiofrequency Thermal Coagulation

- Produces a thermal lesion on the trigeminal nerve. Although pain is relieved, dysesthesia and loss of corneal reflex may occur.

Percutaneous Balloon Micro compression

- Disrupts large myelinated fibers of all 3 branches of the trigeminal nerve.
- Balloon is filled with a contrast material for fluoroscopic identification and compresses the nerve root for 1 minute.



NURSING MANAGEMENT

Preventing Pain

- Providing cotton pads and room-temperature water for washing the face.
- Instruct to rinse with mouthwash after eating when tooth-brushing causes pain, and performing personal hygiene during pain-free intervals.
- Instruct to take food and fluids at room temperature, to chew on the unaffected side, and to ingest soft foods.

Providing Postoperative Care

- Postoperative neurologic assessments
- If the surgery results in sensory deficits to the affected side of the face, the patient is instructed not to rub the eye, because pain will not be felt if there is injury.
- The eye is assessed for irritation or redness.
- Artificial tears may be prescribed to prevent dryness in the affected eye.
- The patient is cautioned not to chew on the affected side until numbness has diminished.
- The patient is observed carefully for any difficulty in eating and swallowing foods of different consistency.

XXIV. MYASTHENIA GRAVIS

- Chronic
- Autoimmune
- Progressive neuromuscular disease affecting the voluntary degree of abnormal weakness and fatigability of skeletal muscles (voluntary muscles)

MANIFESTATIONS

Subjective	Objective
extreme muscle weakness	ptosis of the eyelid
dysphagia	weak voice (dysphonia)
diplopia	myasthenic smile (snarling, nasal smile)
dysarthria (difficulty speaking)	snarling face
dyspnea	Strabismus
	Weakness in intercostal muscles resulting in decreasing vital capacity and respiratory failure

ASSESSMENT AND DIAGNOSTIC FINDINGS

- Acetylcholinesterase Inhibitor Test**
 - Administration of edrophonium chloride (Tensilon)
 - IV; 30 seconds after injection facial muscle weakness and ptosis should resolve for about 5 minutes. Atrophine should be available.
- Serum Assay for Circulating Anti Acetylcholine Receptor**
 - (+) MG if increased
- Ice Test** for pt. with asthma or cardiac condition ice is held over the pt. eye for 1 min. ptosis will be resolved.
- Acetylcholine Antibodies**
- Electromyography**
 - Reduced response to electrical stimulation.

POTENTIAL COMPLICATIONS

Myasthenic Crisis	Cholinergic Crisis
- sudden onset of muscle weakness as a result of under medication or non-cholinergic medication at all - triggered by emotional upset, systemic infection, medication, surgery or trauma.	caused by over medication w/ anticholinesterase
Signs - sudden onset of acute respiratory distress - inability to swallow or speak	Signs - muscle weakness w/ respiratory depression - GI symptoms (nausea, vomiting, diarrhea, increased salivation) - sweating, & bradycardia.

MEDICAL MANAGEMENT

No cure

- Anti cholinesterase**
 - Increase the relative concentration of available Acetylcholine at the neuro muscular junction
 - E.g., Pyridostigmine bromide (Mestinon)
 - Neostigmine bromide (Prostigmin)

2. Immunosuppressive therapy

- Decrease the amount of antibody production.
- E.g., Corticosteroids

3. Cytotoxic Medications

- Decrease the circulating anti acetylcholine receptor
- E.g., Cyclophosphamide (Cytosan)

4. Intravenous Immune Globulin (IVIG)

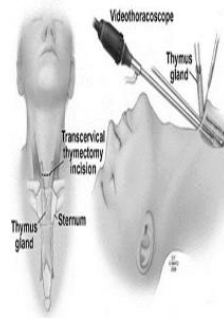
- Treat exacerbation

5. Therapeutic Plasma Exchange (TPE) or plasmapheresis used to treat exacerbations.

SURGICAL MANAGEMENT

Thymectomy

- Can produce antigen-specific immunosuppression.
- It may take 1 year after the removal before the patient can benefit.



NURSING CARE

- Improving respiratory function
- Increasing physical mobility
- Improving communication
- Providing eye care
- Preventing aspiration

DYSFUNCTION OF THE BRAIN

XXV. CEREBROVASCULAR DISORDERS

- Stroke is the primary cerebrovascular disorder in the United States and in the world.
- 2nd leading cause of death.

Types:

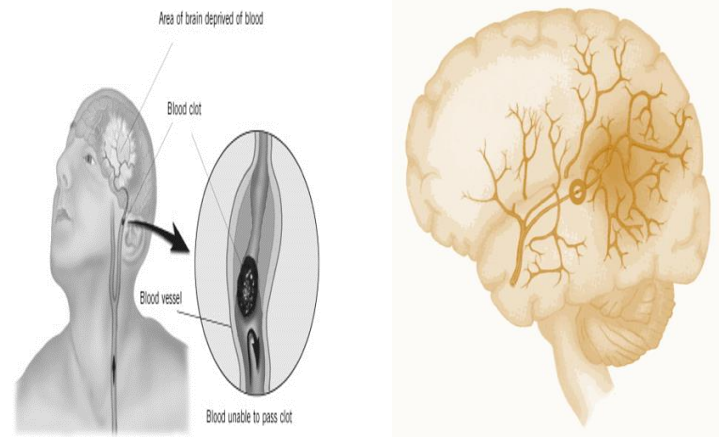
Ischemic – damage to brain due to clogged artery.

Hemorrhagic – blood vessel burst leaking blood into brain tissues.

Table 62-1 • Major Types of Stroke and Their Causes

CLASSIFICATION	CAUSES
Ischemic	Large artery thrombosis Small penetrating artery thrombosis Cardiogenic embolic Cryptogenic (no known cause) Other
Hemorrhagic	Intracerebral hemorrhage Subarachnoid hemorrhage Cerebral aneurysm Arteriovenous malformation

MODIFIABLE RISK FACTORS	NON-MODIFIABLE RISK FACTORS
Hypertension	Age
Hypercholesterolemia	Sex
Atherosclerosis	Family History
Atrial Fibrillation	Past Medical History
Obesity	
Drugs and alcohol	
TIA's	



XXVI. NEUROLOGIC DEFICIT-MANIFESTATION

VISUAL FIELD DEFICITS

1. **Homonymous Hemianopsia (Loss of half of the visual field)**
 - Unaware of persons or objects on side of visual loss.
 - Neglect one side of the body.
 - Difficulty judging distances.
2. **Loss of peripheral vision**
 - Difficulty seeing at night.
 - Unaware of objects or the borders of objects.
3. **Diplopia** – Double vision

MOTOR DEFICITS

1. **Hemiparesis**
 - Weakness of the face, arm, and leg on the same side (due to a lesion in the opposite hemisphere)
2. **Hemiplegia**
 - Paralysis of the face, arm, and leg on the same side (due to a lesion in the opposite hemisphere)
3. **Ataxia**
 - Staggering, unsteady gait.
 - Unable to keep feet together.
 - Needs a broad base to stand

4. **Dysarthria**
 - Difficulty in forming words.
5. **Dysphagia**
 - Difficulty in swallowing.

SENSORY DEFICITS

Paresthesia (occurs on the side opposite the lesion)

- Numbness and tingling of extremity
- Difficulty with proprioception

VERBAL DEFICITS

1. **Expressive Aphasia**
 - Unable to form words that are understandable; may be able to speak in single-word responses.
2. **Receptive Aphasia**
 - Unable to comprehend the spoken word; can speak but may not make sense
3. **Global (mixed) Aphasia**
 - Combination of both receptive and expressive aphasia.

COGNITIVE DEFICITS

- Short- and long-term memory loss
- Decreased attention span
- Impaired ability to concentrate
- Poor abstract reasoning
- Altered judgment

EMOTIONAL DEFICITS

- Loss of self-control
- Emotional lability
- Decreased tolerance to stressful situations
- Depression
- Withdrawal
- Fear, hostility, and anger.
- Feelings of isolation

Table 62-3 • Comparison of Left and Right Hemispheric Strokes

LEFT HEMISPHERIC STROKE	RIGHT HEMISPHERIC STROKE
Paralysis or weakness on right side of body	Paralysis or weakness on left side of body
Right visual field deficit	Left visual field deficit
Aphasia (expressive, receptive, or global)	Spatial-perceptual deficits
Altered intellectual ability	Increased distractibility
Slow, cautious behavior	Impulsive behavior and poor judgment
	Lack of awareness of deficits

XXVII. TIME COURSE IN STROKE

TIA	• Temporary episodes of neurologic dysfunction lasting for seconds or minimum but not longer than 24 hrs. • Complete recovery occurs between attacks
Reversible Ischemic Neurologic Deficit	• Consistent but more pronounce than TIA and last more than 24hrs.
Stroke in Evolution	• Progressing stroke
Completed Stroke	• No further progression of hypoxic insult

DIAGNOSTIC TESTS

- Computed Tomography
- 12 lead electrocardiogram and carotid ultrasound
- Cerebral angiography
- Transcranial Doppler flow studies
- MRI

XXVIII. MEDICAL AND SURGICAL MANAGEMENT

MEDICATIONS

1. Thrombolytic Therapy – Recombinant t-PA

- Minimum dose is 0.9 mg/kg; the maximum dose is 90 mg.
- The loading dose is 10% of the calculated dose and is administered over 1 minute.
- The remaining dose is administered over 1 hour via an infusion pump.
- After the infusion is completed, the line is flushed with 20 mL of normal saline solution to ensure that all the medication is administered.

Side Effects:

- **Bleeding** – most common side effect of t-PA

NURSING RESPONSIBILITIES

- Cardiac monitoring
 - Vital signs are obtained every 15mins for the first 2 hours, every 30mins for the next 6 hours, then every hour for 16 hours.
 - Blood pressure should be maintained with the systolic pressure less than 180 mm Hg and the diastolic pressure less than 100 mm Hg.
 - Airway management is instituted based on the patient's clinical condition and arterial blood gas values.
2. **Anticoagulant** (IV heparin or low-molecular weight heparin) for ischemic strokes.
 3. Careful maintenance of cerebral hemodynamics to maintain cerebral perfusion.
 4. Reduce ICP by administering an osmotic diuretic (eg, mannitol), maintaining PaCO₂ within the range of 35 to 45 mm Hg, and positioning to avoid hypoxia.

- Elevation of the head of the bed to promote venous drainage and to lower increased ICP.
- Intubation with an endotracheal tube to establish a patent airway; if necessary.
- Continuous hemodynamic monitoring.
- Systolic pressure should be maintained at less than 180 mmHg, diastolic pressure at less than 100 mm Hg to reduce the potential for additional bleeding or further ischemic damage.
- Neurologic assessment.

CAROTID ENDARTERECTOMY (CEA)

- Surgical reopening of an artery obstructed by ATHEROMA.
- Removal of an atherosclerotic plaque or thrombus from the carotid.

CAROTID ARTERY STENTING (CAS)

- With or without angioplasty
- Less invasive
- Used for the treatment of carotid stenosis

NURSING MANAGEMENT

- Improving mobility and preventing joint deformity
- Changing positions, maintaining skin integrity
- Establishing exercise program
- Enhancing Self-care, activities of daily living
- Managing Sensory-Perceptual Difficulties
- Attaining bowel and bladder control
- Improving thought process
- Improving communication
- Improving family coping
- Helping pt cope with sexual dysfunction
- Continuing care

XXIX. HEMORRHAGIC STROKE

Causes:

- Intracerebral hemorrhage
- Subarachnoid hemorrhage
- Cerebral aneurysm
- Arteriovenous malformation

Neurologic deficits similar to ischemic stroke plus...

- Sudden, unusually severe headache
- Loss of consciousness
- Nuchal rigidity
- Visual disturbances
- Tinnitus, dizziness
- Hemiparesis

SURGICAL AND MEDICAL MANAGEMENT

- Analgesics
- Bed rest with sedation
- Surgical evacuation (Craniotomy)
 - If cerebellar hemorrhage diameter exceeds 3cm
 - Glasgow Coma Scale is below 14
- Aneurysm clipping
- Craniectomy

XXX. CRANIOTOMY PROCEDURE

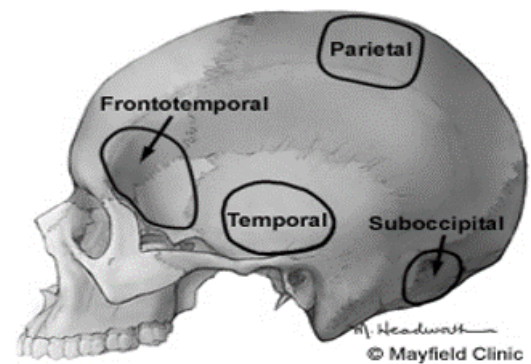


Figure 1. Craniotomies are often named for the bone being removed. Some common craniotomies include frontotemporal, parietal, temporal, and suboccipital.

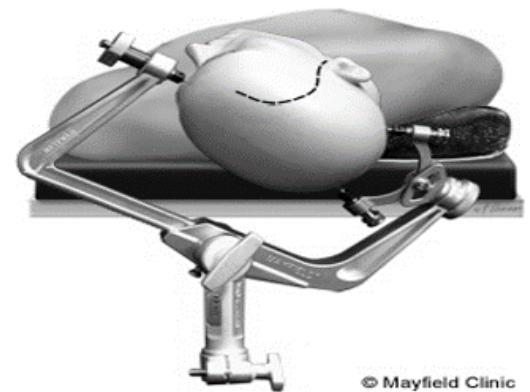


Figure 2. The patient's head is placed in a three-pin Mayfield skull clamp. The clamp attaches to the operative table and holds the head absolutely still during delicate brain surgery. The skin incision is usually made behind the hairline (dashed line).

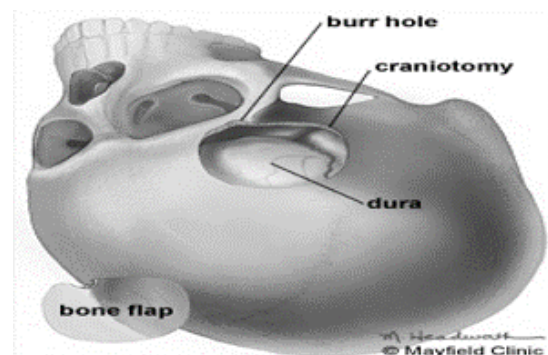


Figure 3. A craniotomy is cut with a special saw called a craniotome. The bone flap is removed to reveal the protective covering of the brain called the dura.

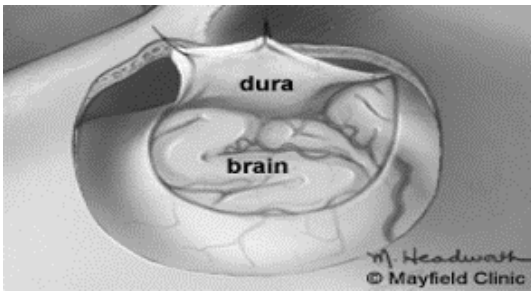


Figure 4. The dura is opened and folded back to expose the brain.

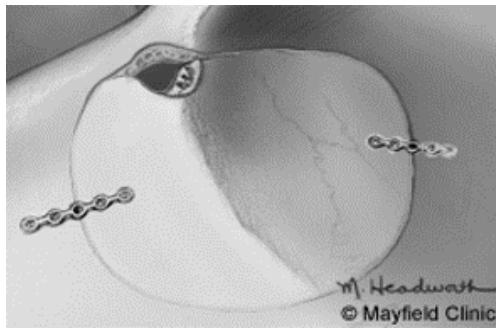


Figure 5. The bone flap is replaced and secured to the skull with tiny plates and screws.

XXXI. ANEURYSM CLIPPING



Figure 1. Most aneurysms resemble a balloon, with a narrow neck at its origin and a large expanding dome. Typically, a clip is placed across the neck of the aneurysm to prevent blood from entering the dome.



Figure 2. Aneurysm clips come in a variety of sizes and curves. A clip applicator opens the blades of the aneurysm clip. They are made of titanium, they stay forever.

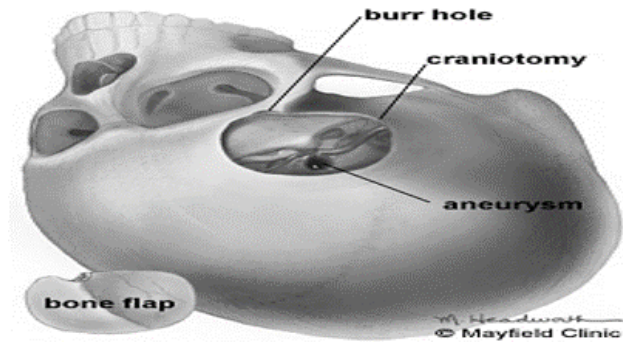


Figure 3. A craniotomy is made in the skull over the area where the aneurysm is located. The bone flap is lifted and temporarily removed.



Figure 4. A titanium clip is placed across the neck of an aneurysm preventing blood from entering.

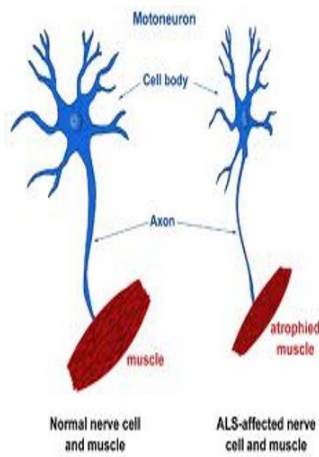
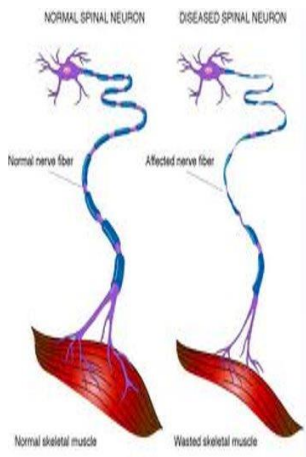
NURSING MANAGEMENT

- Monitor for neurologic deterioration
- Implement aneurysm precaution
- Monitor and manage complications:
 - Vasospasm
 - Seizure
 - Hydrocephalus
 - Rebleeding
- Teaching patient self-care
- Continuing care

DYSFUNCTION OF THE PERIPHERAL NERVOUS SYSTEM

XXXII. AMYOTROPHIC LATERAL SCLEROSIS

- Lou Gehrig Disease
- Is a disease of unknown cause in which there is a loss of motor neurons (nerve cells controlling muscles) in the anterior horns of the spinal cord and the motor nuclei of the lower brain stem.
 - Several theories exist regarding the cause of ALS, including autoimmune disease and free radical damage.
- The leading theory held by researchers is that overexcitation of nerve cells by the neurotransmitter glutamate leads to cell injury and neuronal degeneration
- ALS is incurable and fatal, with median survival of 3 years.



RISK FACTORS

- Smoking
- Viral infections
- Autoimmune disease
- Environmental exposure to toxins
- Common in male between 40-60 years old

SIGNS AND SYMPTOMS

- In 75-80% of patients, symptoms begin with limb involvement.

Initial complaints in patients with lower limb onset are often as follows:

- Tripping, stumbling, or awkwardness when running.
- Foot drop; patients may report a "slapping" gait.

Initial complaints with upper limb onset include the following:

- Reduced finger dexterity, cramping, stiffness, and weakness or wasting of intrinsic hand muscles
- Wrist drop interfering with work performance

With bulbar onset symptoms 1st appear in the face or neck (20-25%), initial complaints are as follows:

- Slurred speech, hoarseness, or decreased volume of speech
- Aspiration or choking during a meal

Emotional and special cognitive difficulties in some ALS patients are as follows:

- Involuntary laughing or crying
- Depression
- Impaired executive function
- Maladaptive social behavior

Features of more-advanced disease are as follows:

- Muscle atrophy becomes more apparent
- Spasticity may compromise gait and manual dexterity
- Muscle cramps are common
- Rarely, painful joint contractures may result from immobility

Progression of bulbar disease leads to the following:

- Voice changes: Hypernasality and development of a strained, strangled vocal quality; eventually, speech may be lost

- Swallowing difficulties, usually starting with liquids
- Drooling

CLINICAL MANIFESTATIONS

- Fatigue
- progressive weakness and atrophy of the muscles of the arms, trunk, or legs, cramps,
- Muscle fasciculations (twitching)
- Muscle incoordination
- Muscle spasticity
- Brisk and overactive deep tendon stretch
- Reflexes
- Difficulty talking, swallowing, and ultimately breathing.
- Weakness of the posterior tongue and palate impairs the ability to laugh, cough, or even blow the nose
- The voice assumes a nasal sound, and articulation becomes so disrupted that the speech is unintelligible.
- Some emotional lability may be present, but intellectual function is not impaired.

MEDICAL MANAGEMENT

- Riluzole (Rilutek), a glutamate antagonist, slows the deterioration of motor neurons (dose of 100 mg riluzole per day)
- Symptomatic treatment and rehabilitative measures are employed to support the patient and improve the quality of life.
- Baclofen (Lioresal), dantrolene sodium (Dantrium), or diazepam (Valium) may be useful for patients troubled by spasticity, which causes pain and interferes with self-care.
- A patient experiencing problems with aspiration and swallowing may require enteral feeding.
- Mechanical ventilation (using negative-pressure ventilators) is an option when alveolar hypoventilation develops.

XXXIII. BELL'S PALSY

- Facial paralysis that occurs on one side of the face as a result of inflamed 7th cranial nerve
- Facial neuropathy characterized by paresis and pain
- Affects men and women equally

The cause is unknown, although possible causes may include **vascular ischemia, viral disease (herpes simplex, herpes zoster), autoimmune disease, or a combination of all of these factors.**

Subjective	Objective
Facial pain Difficulty eating Altered taste	Distortion of face Difficulty with articulation Diminished blink reflex Increased lacrimation

Assessment Findings:

- Presence or absence of blink reflex and ability to close eye
- Facial pain
- Nutritional intake and ability to chew and swallow
- Extent of facial paralysis

MANAGEMENT

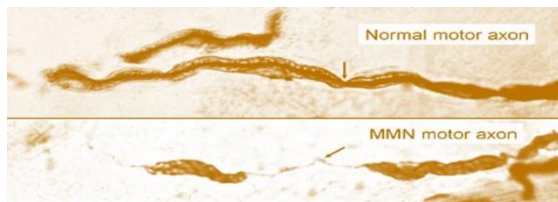
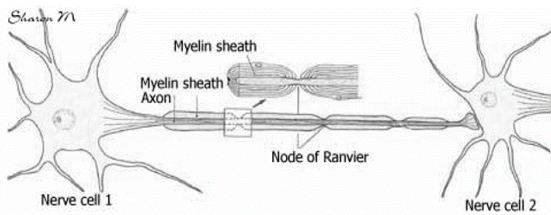
- Recovery 3-5 weeks
- Prednisone therapy
- Analgesic agents
- Heat, massage and electric stimulation
- Prevention of corneal irritation

NURSING CONSIDERATIONS

- Teach client to prevent corneal irritation by using artificial tears, manually closing the eye & applying an eye protective shield.
- Teach client to keep the face warm.
- Teach the client to massage face gently & perform simple facial exercises such as wrinkling, blowing, whistling.
- Encourage ventilation of feelings because self-image may be affected.
- Support client's nutritional status by providing privacy & small, frequent feedings & encourage the client to favor the unaffected side while eating.
- Avoid exposure of the face to colds.

XXXIV. GUILLAIN-BARRE SYNDROME (ACUTE IDIOPATHIC POLYNEURITIS)

- An autoimmune attack of the peripheral nerve myelin
- Characterized by acute neuropathy involving motor and sensory nerves throughout the peripheral nervous system.

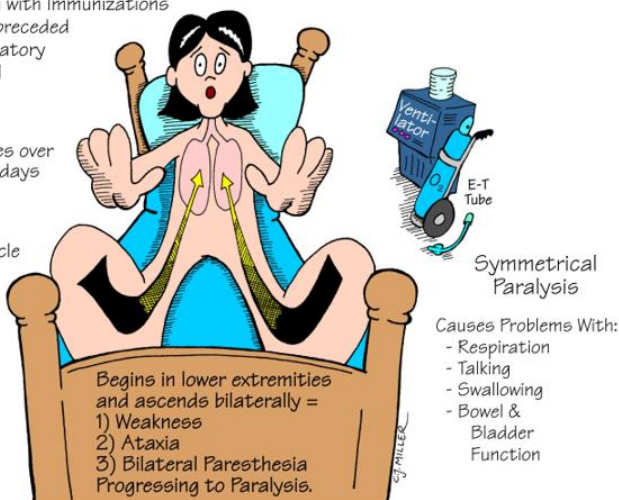


Risk Factors:

- Possibly Autoimmune
- Association with Immunizations
- Frequently preceded by mild respiratory or intestinal infection

- Progresses over hours to days

- Minimal Muscle Atrophy



TYPES

1. Ascending GBS Weakness in the lower extremities which progresses upward and has potential for respiratory failure
2. Purely motor with no altered sensation
3. Descending GBS affects the head and neck
4. Miller-Fisher variant rarest type (ataxia, areflexia and ophthalmoplegia)

Assessment findings must include	Diagnostic tests
Occupation Infectious and exposure history Extension of weakness Location and movement and progression of paresis Vital signs Signs of decreased oxygenation	Exclusion of other symptoms (Poliomyelitis, botulism, exposure to heavy metals) History of the patient Muscle strength tests Muscle activity tests Physical examination Reflex tests, such as the knee-jerk reaction Nerve conduction velocity tests Spinal tap, to check for higher than expected levels of protein in the cerebrospinal fluid. CSF analysis

CLINICAL MANIFESTATIONS

- Muscle weakness
- Diminished reflexes
- Hyporeflexia
- Neuromuscular respiratory failure
- Paresthesia
- Blindness
- Bulbar muscle weakness
- Areflexia

MEDICAL MANAGEMENT

- **Plasmapheresis (IVIG)** - trials have proven the effectiveness of this form of treatment.
- Support of vital function: NGT insertion, intermittent catheterization, intubation or ventilatory support.
- Anticoagulant agent
 - Steroids
 - IVF therapy

NURSING MANAGEMENT

- Monitor vital signs, vital capacity, breath sounds & ABG
- Keep airway & tracheostomy set at bedside
- Suction, provide fluid replacement therapy, & monitor functioning of the respirator as required
- Provide emotional support to the client & family
- Provide explanations of disease process & care
- Refer client & family to a support group or foundation for additional information & resources
- Prevent complications of Immobility
- Skin care, apply antiembolic stockings
- Range of motion exercise, position changes
- Coughing & deep breathing